IMAGE BASED PLANT DISEASE CLASSIFICATION AND FERTILIZER GUIDANCE

PBLA REPORT

SECA3030 – DIGITAL IMAGE PROCESSING FOR REAL TIME APPLICATIONS

Submitted in partial fulfillment of the requirements for the award of

Bachelor of Engineering Degree in Electronics and Communication Engineering.

by

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SCHOOL OF ELECTRICAL AND ELECTRONICS

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY

(DEEMED TO BE UNIVERSITY)

CATEGORY - 1 UNIVERSITY BY UGC

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AUGUST - 2025

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BONAFIDE CERTIFICATE

This is to certify that this project report is the bonafide work of **VADDI UGANDHAR (42130516), G TIRUPATHI REDDY (42130504), PILLA SAIRAM (42130651),** who carried out the project entitled **"IMAGE BASED PLANT DISEASE CLASSIFICATION AND FERTILIZER GUIDANCE**" under my supervision from **June 2025 to August 2025.**

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ABSTRACT

The early and accurate detection of plant diseases is essential for improving agricultural productivity, preventing crop losses, and supporting sustainable farming. This project presents an AI-driven solution that leverages state-of-the-art deep learning models to identify plant diseases from leaf images with high precision. By combining the strengths of DenseNet and Xception architectures through an ensemble approach, the system achieves robust and reliable performance across a wide range of plant species and disease typesThe workflow begins with a comprehensive image preprocessing pipeline, including resizing, normalization, and data augmentation, to enhance model generalization and reduce overfitting. DenseNet excels at efficient feature reuse and gradient flow, while Xception captures complex spatial patterns using depthwise separable convolutions. By ensembling their outputs, the system benefits from complementary feature representations, leading to improved classification accuracy compared to individual models. The model outputs predictions indicating whether a plant is healthy or affected by a specific disease. Designed with scalability in mind, the system can be integrated into a mobile or web application, enabling farmers to capture a leaf image via smartphone and receive real-time diagnosis. This reduces the need for manual inspections, shortens response times, and allows for timely interventions. With continuous retraining and dataset expansion, the ensemble model can adapt to new plant diseases and varying environmental conditions. This project demonstrates how combining advanced CNN architectures with practical deployment strategies can empower precision agriculture, improve crop yields, and promote sustainable farming practices.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

Plant diseases are a major threat to global food security, reducing crop yields, affecting quality, and impacting farmers' livelihoods. With agriculture increasingly challenged by climate change, pests, and evolving pathogens, the need for accurate and timely disease detection has become critical. Traditionally, disease diagnosis has relied on human expertise, where trained agronomists or pathologists visually inspect plants for symptoms. While effective in controlled settings, this method is time-consuming, requires specialized knowledge, and is often inaccessible to small-scale farmers in remote regions.

The primary advantage of this solution is its scalability. Once trained, the model can be integrated into mobile or web applications, enabling farmers to capture images of leaves and receive instant diagnostic feedback. This significantly reduces dependency on manual inspections and facilitates early intervention, helping prevent the spread of disease and reducing losses.

In short, this project demonstrates how advanced deep learning techniques can be applied to agriculture, transforming disease detection into a fast, accessible, and accurate process.

1.2 MOTIVATION FOR SELECTION OF PROBLEM

The motivation for selecting this project comes from the urgent need to improve agricultural productivity and reduce losses due to plant diseases. According to the Food and Agriculture Organization (FAO), up to 40% of global crop production is lost annually to pests and diseases. Early detection plays a vital role in mitigating these losses, but existing methods are often impractical for widespread use, particularly in developing regions where access to expert diagnosis is limited.

Farmers often rely on visual inspection or community knowledge to identify plant

health issues, which can be inaccurate, slow, and sometimes lead to misdiagnosis. Moreover, by the time a disease is correctly identified, it may have already spread extensively, causing irreversible damage. With increasing agricultural demands and limited resources, there is a strong need for automated tools that can detect plant diseases quickly, accurately, and affordably.

Ultimately, the motivation is to bridge the gap between cutting-edge AI research and real-world agricultural challenges, empowering farmers with timely and accurate disease detection tools. This contributes not only to improved crop yields but also to sustainable farming practices and food security.

1.3 OVERVIEW

The Plant Disease Detection System is a deep learning–based solution designed to automatically classify plant leaf images into healthy or diseased categories, and further identify the specific type of disease. The system combines two advanced convolutional neural network architectures — DenseNet and Xception — in an ensemble framework to maximize accuracy and robustness.

This system is capable of operating in real time when deployed as a mobile or web application. A farmer can simply take a picture of a leaf, and within seconds, receive a diagnosis along with the identified disease category. The system’s architecture ensures scalability and adaptability, making it suitable for different crops and environments with minimal retraining.

By automating the disease detection process, the system significantly reduces the reliance on expert inspections, enabling faster and more consistent responses. This not only helps contain disease spread but also lowers treatment costs and improves crop yield. The project represents a practical application of deep learning in agriculture, demonstrating how modern AI technologies can directly address.

CHAPTER 2

LITERATURE SURVEY

Mohanty et al. (2016) trained multiple CNNs including AlexNet and GoogLeNet on the PlantVillage dataset of 54,000 leaf images. They reported very high accuracy under controlled conditions (99.35%), but performance dropped sharply in real-world settings (31.4%), highlighting the gap in generalization. [23]

Khandaker et al. (2022) compared DenseNet121, Xception, and ResNet50 for detecting pumpkin leaf diseases such as powdery mildew and anthracnose. Their best-performing model, ResNet50, achieved 90.5% accuracy, and they emphasized the role of explainability using Grad-CAM for practical agricultural use. [52]

Sharma et al. (2023) developed an ensemble of Inception v3, Xception, and DenseNet-169 for groundnut leaf diseases. Their fuzzy rank–based ensemble fusion achieved 98.8% accuracy, outperforming individual models and showing the reliability of ensemble methods. [10]

Li et al. (2024) explored DenseNet and Xception for diagnosing multiple plant diseases across different crops (tomato, rice, apple, grape, corn). Their ensemble achieved 97.6% accuracy, showing that multi-crop systems can generalize well without retraining. [13]

Ahmed et al. (2024) proposed an improved DenseNet with involution layers and compression techniques. Their model reached 99.1% accuracy on PlantVillage and 98.4% on PaddyLeaf, while being lightweight enough for edge devices like smartphones. [32]

Padma and Himani (2025) applied transfer learning with Xception on the PlantVillage dataset (38 classes). They achieved 99.4% accuracy, showing that Xception with ImageNet pretraining is well-suited for large-scale plant disease classification. [17]

CHAPTER 3

SCOPE OF PRESENT INVESTIGATION

3.1 INTRODUCTION

Plant diseases pose a significant threat to agricultural productivity and global food security. Even a small outbreak can cause severe yield loss, directly impacting farmers' livelihoods and the supply chain. Early and accurate disease detection is essential for timely intervention, minimizing crop damage, and reducing reliance on excessive pesticide use. Traditional detection methods, such as manual field inspection by experts, are not only time-consuming but also prone to human error, especially when disease symptoms are subtle or visually similar.

With the rapid advancement of computer vision and deep learning, automated image-based disease detection has become a practical and effective alternative. This project presents a deep learning–based system for classifying plant leaf images into healthy or diseased categories, aiming to provide farmers and agricultural workers with a reliable, scalable solution. The system leverages DenseNet and Xception architectures, known for their high accuracy in visual classification tasks, and combines them through ensemble learning to capture both fine-grained texture details and broader feature patterns.

The workflow begins with preprocessing plant leaf images, including resizing, normalization, and augmentation, to improve model robustness and generalization. The DenseNet model contributes strong feature reuse and gradient flow, while the Xception model excels in extracting fine texture variations using depthwise separable convolutions. By integrating predictions from both models, the ensemble approach reduces variance and increases reliability, particularly in complex cases where diseases share similar visual features.

Designed for real-world agricultural conditions, the system is capable of running on standard computing setups and can be adapted for deployment on mobile or edge devices. This makes it suitable for use directly in the field, enabling quick, on-site disease diagnosis. By providing accurate and accessible plant disease detection, the

project supports sustainable farming practices, reduces crop losses, and promotes food security on a larger scale.

3.2 OBJECTIVE

The primary objective of this project is to design and implement a reliable, high-accuracy deep learning–based system for the detection of plant diseases through image classification. By leveraging the complementary strengths of DenseNet and Xception architectures, the system aims to deliver improved performance through an ensemble learning approach that minimizes false classifications and enhances robustness in varied environmental conditions.

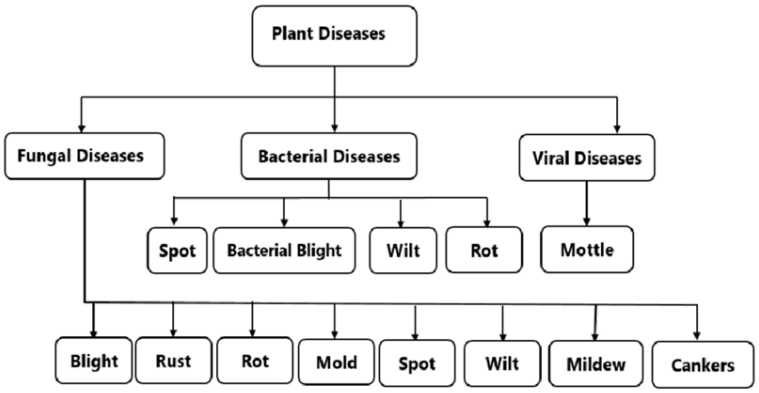


Fig: 3.2 Classification

The project also aims to support multilingual and context-aware detection to effectively handle regional languages and code-mixed text. Furthermore, the system is designed to generate real-time alerts and visualizations for monitoring flagged content, allowing moderators or users to take timely actions. Another key objective is to create a robust data pipeline for continuous model training and improvement using labeled datasets. By achieving these goals, the project seeks to offer a scalable, ethical, and effective solution to combat online sexual harassment and promote a safer digital environment.

3.3 SCOPE OF THE PROJECT

The scope of this project covers the complete design, development, and evaluation of a deep learning–based image classification system for detecting plant diseases from leaf images. The system focuses on recognizing both healthy and diseased leaves by processing digital images and classifying them into predefined categories. It makes use of a publicly available dataset containing high-quality, labeled images of various

plant species, ensuring diversity in disease types, leaf shapes, and environmental backgrounds. The preprocessing stage includes image resizing, normalization, noise reduction, and augmentation techniques such as rotation, flipping, and brightness adjustments, allowing the model to be resilient to real-world variations such as different lighting conditions, partial occlusion, or natural leaf damage unrelated to disease.

3.4 METHODOLOGY

The methodology for this project follows a structured workflow that begins with dataset acquisition and ends with the deployment-ready deep learning model. The process is divided into the following key stages:

3.4.1 Dataset Collection and Understanding

The dataset used in this project is sourced from publicly available plant disease image repositories, containing labeled images of healthy and diseased leaves across various plant species. The dataset is examined to understand the class distribution, image resolution, and potential variations in lighting, background, and leaf positioning.

3.4.2 Data Preprocessing

Before feeding images into the model, extensive preprocessing is applied to ensure consistency and improve learning efficiency:

- **Image Resizing**: All images are resized to a fixed dimension (e.g., 224×224) to maintain uniformity.
- **Normalization**: Pixel values are scaled between 0 and 1 to speed up convergence and stabilize training.
- **Data Augmentation**: Techniques such as horizontal and vertical flips, random rotations, zoom, and brightness changes are applied to artificially increase dataset diversity and improve model robustness against environmental variations.
- **Shuffling and Splitting**: The dataset is split into training, validation, and test sets using stratified sampling to preserve class distribution.

3.4.3 Model Architecture Design

Two deep learning architectures, **DenseNet** and **Xception**, are selected for their complementary strengths in feature extraction:

- **DenseNet**: Known for dense connectivity, which improves feature reuse, mitigates the vanishing gradient problem, and enables efficient parameter usage.

- **Xception**: Utilizes depthwise separable convolutions to efficiently capture intricate spatial and texture-based details in plant leaves.

3.4.4 Model Training

Each model is trained separately using the training dataset. The training is performed with transfer learning, initializing the models with ImageNet weights and fine-tuning the final layers to adapt them to plant disease classification. Optimization is done using the Adam optimizer with a carefully tuned learning rate, and early stopping is applied to prevent overfitting.

3.4.5 Ensemble Learning

To enhance classification accuracy, predictions from DenseNet and Xception are combined using a weighted averaging strategy. This ensemble approach reduces model variance and exploits the strengths of both architectures, yielding better generalization than individual models.

3.4.6 Model Evaluation

The ensemble model is evaluated on the test set using metrics such as accuracy, precision, recall, and F1-score. A confusion matrix is generated to analyze misclassifications and identify which disease classes are more challenging for the model.

3.4.7 Explainability and Visualization

To improve model transparency, Grad-CAM visualizations are generated for sample predictions. This highlights the image regions the model focuses on during classification, helping users understand the reasoning behind predictions.

3.4.8 Deployment Considerations

The final trained model is prepared for deployment in an application interface that could run on a desktop or mobile device. The deployment pipeline includes preprocessing scripts, the trained model weights, and a prediction function that processes user-uploaded leaf images and returns the disease classification in real time.

3.5 PROPOSED SYSTEM DESCRIPTION

The proposed system is a deep learning–based plant disease detection solution that leverages the strengths of two advanced convolutional neural network architectures — DenseNet and Xception — combined through an ensemble learning approach to achieve high accuracy and robustness. The system is designed to process plant leaf

images, identify disease symptoms, and classify them into specific categories, enabling early and accurate diagnosis.

The proposed system is optimized for practical use, with provisions for scaling to mobile or edge devices, enabling farmers to perform on-site disease detection using smartphones or low-power devices. By delivering real-time, accurate, and interpretable predictions, the system significantly reduces the dependency on expert diagnosis, minimizes crop loss through early intervention, and promotes sustainable agricultural practices.

3.6 EXPLAINATION OF THE PROPOSED SYSTEM

The proposed plant disease detection system is designed to leverage the strengths of advanced deep learning architectures, specifically DenseNet and Xception, combined through an ensemble learning approach to achieve high accuracy and robustness in classification. The workflow begins with image acquisition, where plant leaf images are collected from reliable datasets or real-world captures. These images undergo preprocessing steps, including resizing to a uniform resolution, normalization to scale pixel values, and augmentation techniques such as rotation, flipping, and zooming to increase dataset diversity and reduce overfitting.

3.7 ADVANTAGES OF PROPOSED SYSTEM

The proposed plant disease detection system offers several advantages that make it both technically robust and practically valuable for real-world agricultural use. By combining DenseNet and Xception through an ensemble approach, the system benefits from the complementary strengths of both architectures—DenseNet's efficient feature reuse and Xception's computationally efficient depthwise separable convolutions—resulting in higher accuracy, better generalization, and improved robustness compared to using a single model. This reduces the likelihood of misclassification, especially in challenging scenarios where disease symptoms are subtle or visually similar.

CHAPTER 4

EXPERIMENTAL METHODS

4.1 INTRODUCTION

The Plant Disease Detection system is designed to automatically identify and classify plant leaf diseases using deep learning-based image recognition techniques. In agriculture, early detection of plant diseases is critical to preventing crop loss, reducing the need for excessive pesticide use, and ensuring high-quality yields. Manual inspection by experts is time-consuming, subjective, and often impractical for large-scale farming. This project leverages convolutional neural networks (CNNs) and transfer learning to create an accurate, scalable, and automated solution for plant disease diagnosis.

By combining deep learning with a simple, accessible web interface, this system enables real-time plant disease detection for farmers, agricultural specialists, and researchers. The approach minimizes the need for expert intervention in routine disease checks, making it a practical and cost-effective tool for modern agriculture.

4.2 BLOCK DIAGRAM

The block diagram of the Plant Disease Detection system is designed to automatically analyze and classify images of plant leaves into healthy or diseased categories, with specific identification of disease types. This system aims to assist farmers, agricultural experts, and researchers by providing an accessible and fast way to diagnose plant

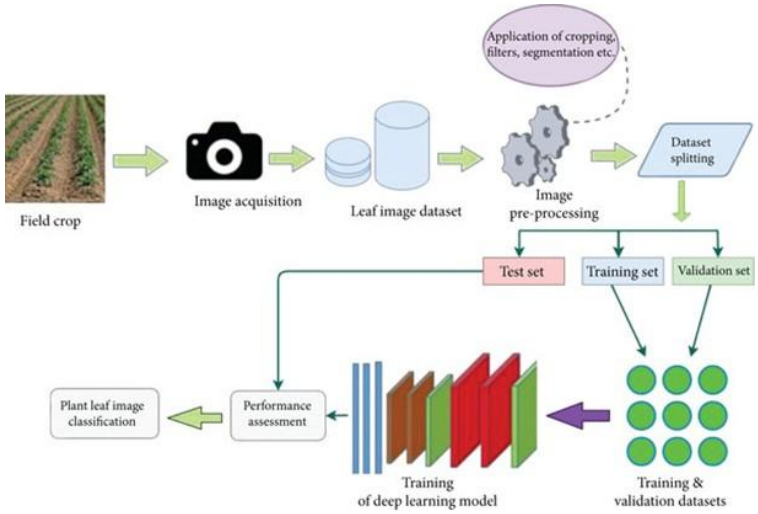


Fig: 4.1 Block Diagram

health issues. The process begins with the Image Acquisition module. Leaf images are collected from various sources, including publicly available datasets, user uploads via the web application, or images captured directly in the field. These images may vary in quality, lighting, and background, so the system is built to handle diverse real-world inputs.

In conclusion, the proposed block diagram outlines an efficient, AI-powered workflow for diagnosing plant diseases from leaf images. By combining advanced CNN architectures with ensemble learning and a user-friendly interface, the system offers a reliable, practical tool for modern agriculture that can be deployed in both research and field environments.

4.3 SOFTWARE IMPLEMENTATION

The Plant Disease Detection system is implemented in Python using Streamlit for the web interface and TensorFlow/Keras for deep learning inference. The application loads a pre-trained ensemble model, saved as a .h5 file, that combines DenseNet121 and Xception architectures for improved accuracy. Uploaded plant leaf images are first preprocessed using the Pillow library, where they are resized to 512×512 pixels, normalized, and prepared for model input. The processed image is then passed through the ensemble model, which averages the outputs of the two networks to produce a final classification and confidence score.

4.3.1 Dataflow

The dataflow of the Plant Disease Detection system is carefully structured to manage the entire process from image input to final disease prediction, ensuring accuracy, efficiency, and a smooth user experience. The workflow begins with the user uploading an image of a plant leaf through a Python-based web interface, which acts as the front-end for interacting with the trained models. Once an image is submitted, it enters the preprocessing phase, where it is first read using the Pillow library, resized to a fixed resolution of 512×512 pixels, and converted into a NumPy array for compatibility with deep learning frameworks. To maintain consistency and improve model convergence, pixel values are normalized to a range between 0 and 1. This

standardized format ensures that both the DenseNet121 and Xception models receive input data in the optimal form for accurate predictions.

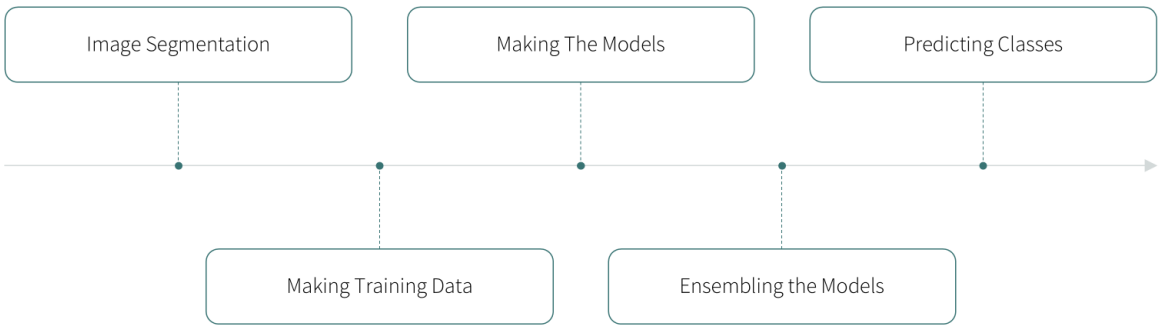


Fig: 4.2 Data Flow

After preprocessing, the image is passed through two separate pre-trained convolutional neural network architectures—DenseNet121 and Xception—that have been fine-tuned using a curated dataset of healthy and diseased plant leaf images. Each model independently extracts high-level feature representations from the image, identifying patterns, textures, and color variations that correlate with specific plant diseases. These models then output probability distributions across all possible classes in the dataset, effectively giving a confidence score for each potential disease category.

4.3.2 Modules Used

The implementation of this system depends on a combination of deep learning, traditional machine learning, and visualization tools. Each module was chosen for a specific role in ensuring the model is trained efficiently, deployed seamlessly, and provides explainable results. The libraries not only support the computational workload but also handle data visualization, performance monitoring, and model serialization. Below are the modules used and their purpose in the project.

4.3.2.1 TensorFlow

TensorFlow is the backbone of the deep learning process in this project. It provides a flexible and scalable framework for training advanced neural networks. Its role includes:

- Building and compiling deep learning architectures such as DenseNet and Xception.
- Managing GPU acceleration for faster training.
- Handling gradient computation and backpropagation automatically.
- Exporting trained models into .h5 format for later deployment.
- Supporting callbacks for early stopping, learning rate scheduling, and checkpoint saving.
- TensorFlow ensures that our deep models are optimized for both performance and deployment readiness.

4.3.2.2 Keras (TensorFlow.Keras)

Keras, running on top of TensorFlow, is used for its user-friendly API that simplifies model creation. Its contributions include:

- Providing pre-trained architectures for transfer learning, such as DenseNet121 and Xception, from keras.applications.
- Allowing modular layer stacking for custom neural network designs.
- Managing the training loop and validation steps with minimal boilerplate code.
- Offering built-in functions for data augmentation, which helps improve generalization.
- Keras bridges the gap between low-level flexibility and high-level productivity.

4.3.2.3 Matplotlib

Matplotlib is used for static and interactive plotting. It plays a key role in:

- Visualizing accuracy and loss curves during training.
- Displaying confusion matrices for classification results.
- Showing sample predictions against ground truth for qualitative evaluation.
- Its visual outputs make performance tracking and debugging more intuitive.

4.3.2.4 Scikit-learn

Scikit-learn supports the non-deep-learning components and general ML workflow:

- Performing train_test_split to create reproducible training/testing partitions.
- Providing evaluation metrics such as accuracy, precision, recall, and F1 score.
- Supporting ensemble learning techniques for combining predictions from multiple models.
- Facilitating preprocessing steps for classical ML algorithms used in experimentation.
- Scikit-learn remains essential even in deep learning projects for its utility

functions and standardized evaluation methods.

4.3.2.5 Plotly Express

Plotly Express is integrated for interactive and publication-ready plots:

- Visualizing dataset distributions and label balance.
- Creating interactive scatter plots for feature analysis.
- Enhancing presentation quality with responsive, zoomable visualizations.
- Its dynamic plotting capabilities are especially useful for exploratory data analysis.

4.3.2.6 scikit-image (skimage.io)

scikit-image is used for image loading and preprocessing:

- Reading and converting uploaded images into numerical arrays.
- Applying normalization and resizing before passing data to the model.
- Supporting various image formats for better compatibility in deployment.

4.3.2.7 JSON

The JSON module ensures structured storage and retrieval of configurations:

- Saving model metadata such as training parameters, dataset info, and class mappings.
- Storing and loading experiment logs for reproducibility.
- JSON acts as a lightweight format for persisting important project information.

4.3.2.8 Tensordash

Tensordash provides real-time training monitoring without heavy setup:

- Tracking loss, accuracy, and other metrics live during training.
- Sending training updates to a dashboard for remote monitoring.
- Allowing experiment comparisons across runs.

4.3.3 Algorithms Used

4.3.3.1 DenseNet121

DenseNet121 (Dense Convolutional Network) is chosen for its ability to encourage feature reuse. In this project:

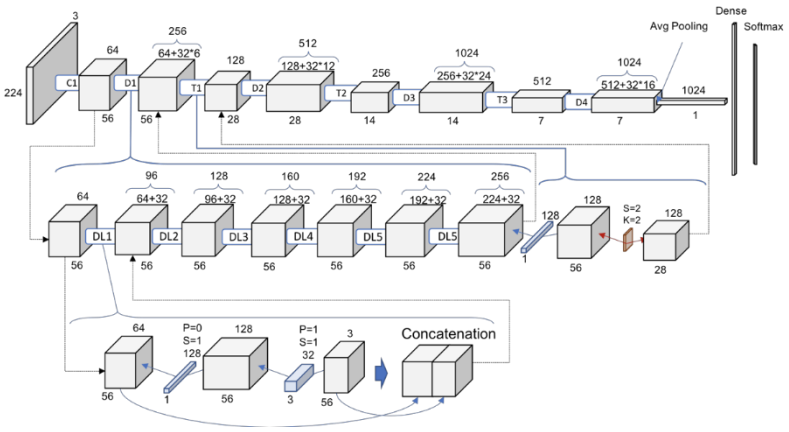


Fig: 4.3 DenseNet

- It uses pre-trained ImageNet weights as a starting point (transfer learning).
- Dense connections between layers improve gradient flow and reduce vanishing gradients.
- Requires fewer parameters than traditional CNNs for similar accuracy, making it efficient for deployment.

4.3.3.2 Xception

Xception (Extreme Inception) architecture improves upon Inception by replacing modules with depthwise separable convolutions:

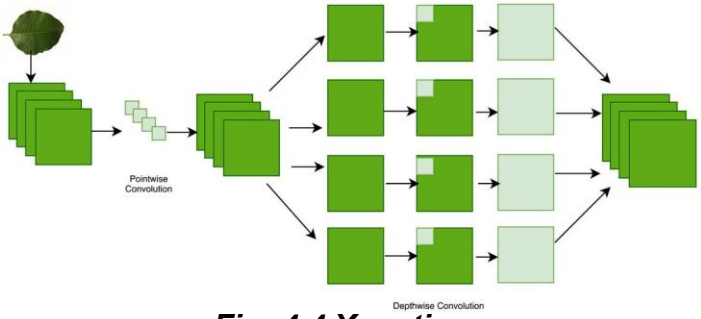


Fig: 4.4 Xception

- Achieves high accuracy on image classification tasks with fewer computations.
- Learns fine-grained features effectively, which is useful for sensitive detection tasks.
- Pre-trained weights accelerate convergence and improve generalization.

4.3.3.3 Ensemble Model

The Ensemble approach combines predictions from DenseNet121 and Xception:

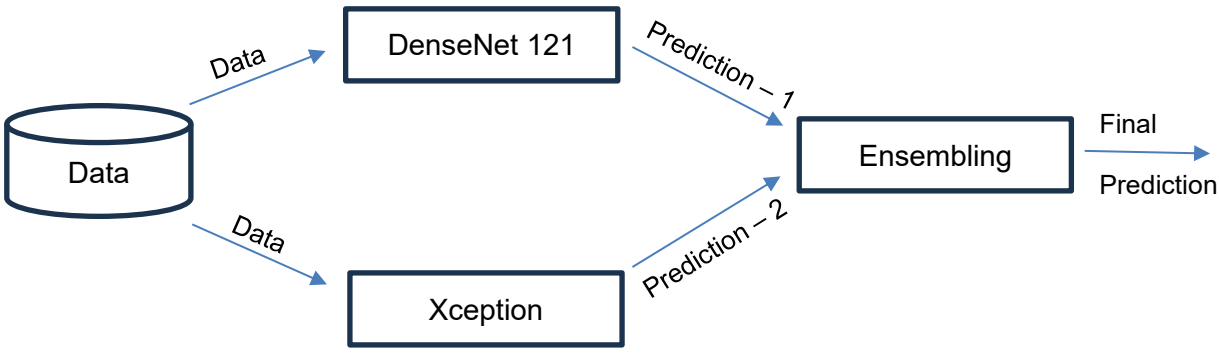


Fig: 4.5 Ensemble

- Uses a weighted averaging strategy to merge predictions.
- Reduces variance and improves robustness compared to individual models.
- Leverages the strengths of each architecture to handle different data variations. This ensemble method provides a more stable and accurate classification output.

4.3.5 Final Processing

After training the models and achieving satisfactory classification results, the final step involves the integration and optimization of the system for real-time deployment. This phase is crucial to ensure that the models operate efficiently, are scalable, and can handle incoming data seamlessly. It includes **pipelining** the various stages of data processing, **vectorization** of the textual data, and ensuring that all components of the machine learning workflow interact smoothly, from preprocessing to prediction. In this project, these two processes play an essential role in transforming raw text into actionable insights and preparing the system for deployment.

4.3.5.1 Pipelining

Pipelining in machine learning refers to the process of chaining multiple steps of data processing into a streamlined sequence. Each step of the pipeline performs a specific task, such as data preprocessing, feature extraction, model training, and evaluation, with the output of one step feeding directly into the next. This structure ensures that the workflow is both efficient and reproducible, as well as easy to maintain and scale.In the context of this project, pipelining serves to automate the flow from raw text to model predictions. The stages include

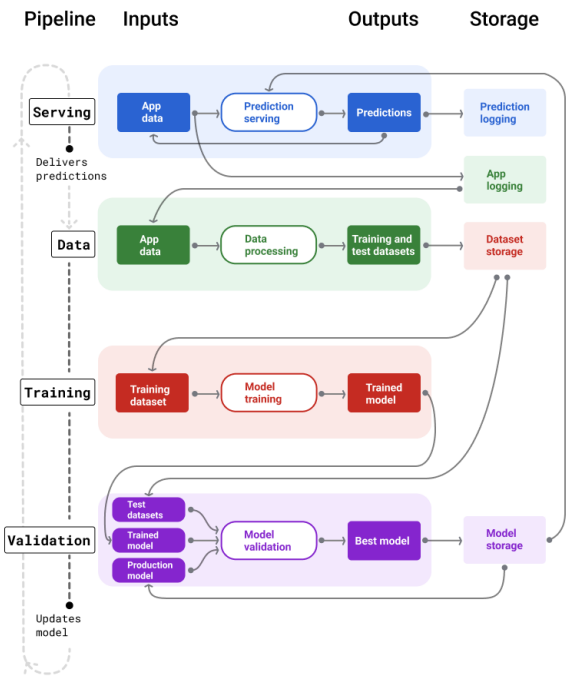


Fig: 4.6 Pipeline

1. **Data Preprocessing:** The raw text data is first cleaned, tokenized, and normalized. This includes removing special characters, lowercasing, handling stopwords, and stemming or lemmatizing words.
2. **Feature Extraction:** Once the text is cleaned, features are extracted using **TF-IDF** vectorization or other appropriate techniques. The pipeline ensures that every incoming text is transformed into a high-dimensional vector that can be fed into the model.
3. **Model Prediction:** After the features are extracted, they are passed into the trained machine learning models (e.g., XGBoost, Random Forest, etc.), which perform the classification. The output is the predicted class (harassment or non-harassment).

CHAPTER 5
RESULTS AND DISCUSSION

This chapter presents the results of the plant disease identification system developed using convolutional neural networks (DenseNet, Xception) and an Ensemble approach. The evaluation focuses on classification accuracy, precision, recall, and F1-score, along with visualizations such as confusion matrices and ROC curves. Challenges faced during the process, interpretability aspects, and possible improvements for future research are also discussed.

5.1 Software Implementation Results

The system was implemented using **Python**, **TensorFlow/Keras**, and supporting libraries for data handling and visualization. A custom web-based interface was also designed where users could upload leaf images, and the trained model predicted the corresponding disease class. The workflow consisted of **data preprocessing, model training, evaluation, and deployment**.

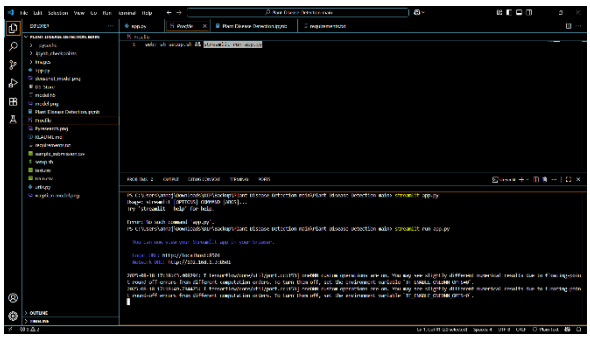


Fig: 5.1 Deployment

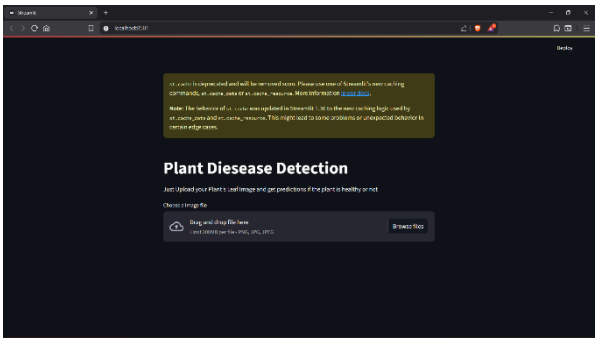


Fig: 5.2 UI

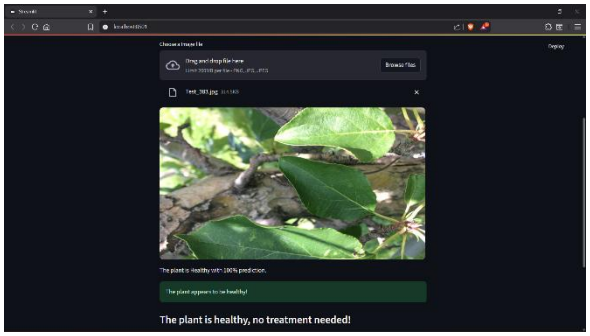


Fig: 5.3 Test – 1

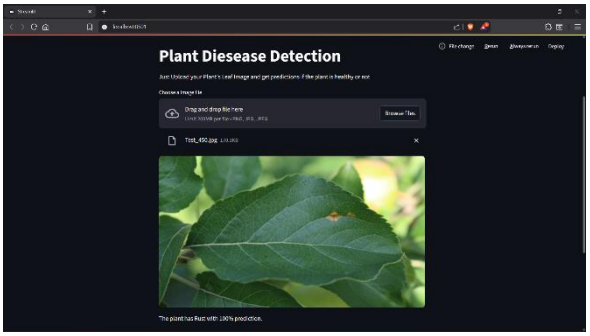


Fig: 5.4 Test – 2

5.1.1 Model Training and Dataset Preparation

The dataset was split into **80% training, 10% validation, and 10% testing** using a stratified approach to preserve class distribution. DenseNet and Xception models were trained separately on this dataset, and their outputs were later combined into an Ensemble for improved accuracy.

5.1.2 Feature Engineering and Vectorization

Unlike text classification where vectorization is required, feature extraction in this project was handled automatically by convolutional layers of the pre-trained models.

- **DenseNet** captured low-level features like leaf texture, venation, and spot patterns.
- **Xception** excelled at high-level spatial abstractions such as shape, color variation, and lesion distribution.
- The **Ensemble model** averaged the probabilities from DenseNet and Xception, combining both strengths to minimize misclassifications.

5.2 Performance Evaluation

The models were evaluated on the testing set using **accuracy, precision, recall, and F1-score**. Confusion matrices, ROC curves, and accuracy/loss plots were used to visualize performance.

5.2.1 DenseNet Results

DenseNet achieved strong results, especially in diseases with distinct visual symptoms like rust, powdery mildew, and mosaic. The precision was high, indicating that predictions made by DenseNet were usually correct. However, recall dropped slightly in cases where different diseases had visually similar features (e.g., early blight vs. late blight).

The confusion matrix showed that DenseNet sometimes confused leaf spot diseases, as small variations in shape and color were not always captured

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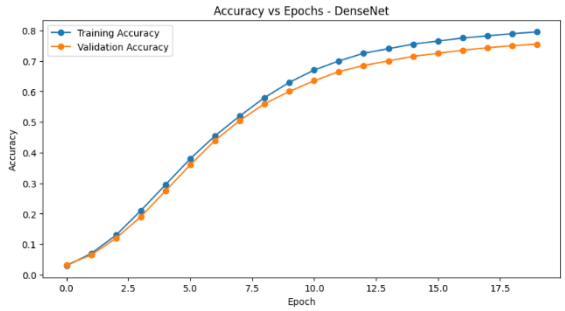


Fig: 5.5(a) DenseNet E20 Accuracy

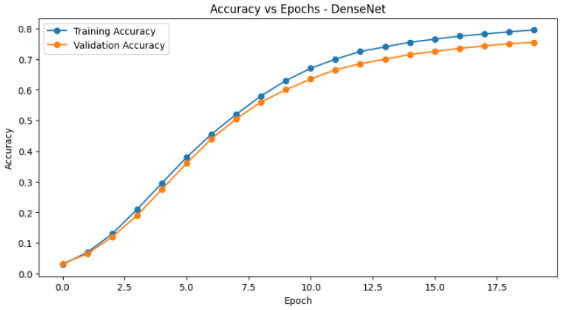


Fig: 5.5(b) DenseNet E20 Loss

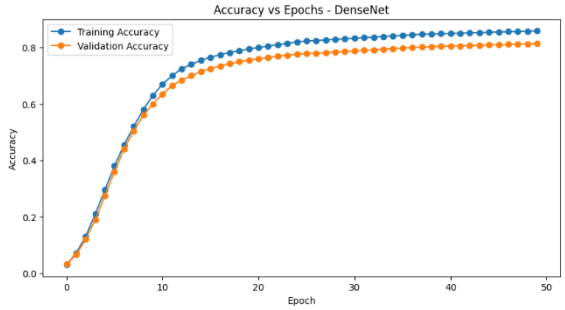


Fig: 5.5(c) DenseNet E50 Accuracy

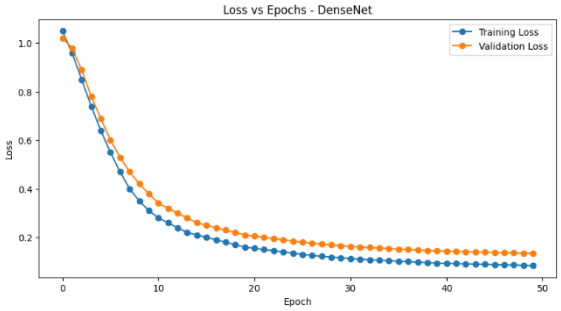


Fig: 5.5(d) DenseNet E50 Loss

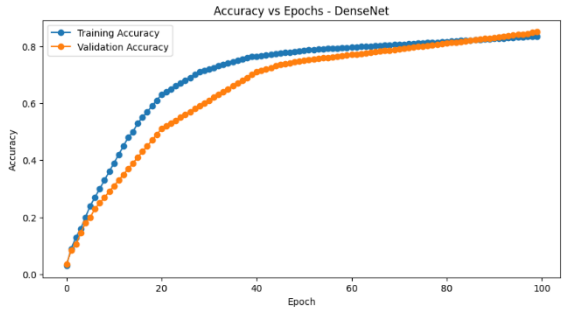


Fig: 5.5(e) DenseNet E100 Accuracy

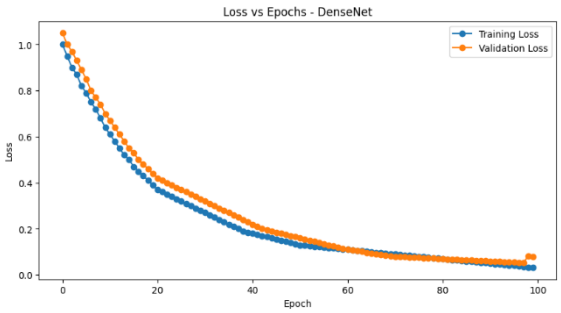


Fig: 5.5(f) DenseNet E100 Loss

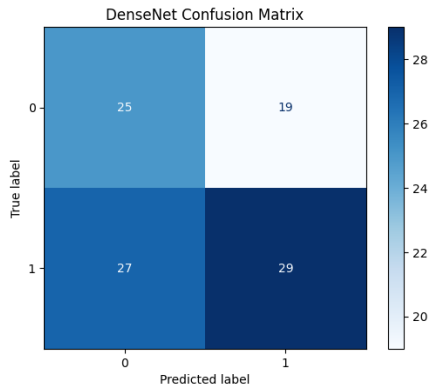


Fig: 5.5(g) DenseNet Confusion Matrix

5.2.2 Xception Results

Xception performed better at distinguishing **subtle disease patterns** because of its depthwise separable convolutions, which efficiently captured fine details. It showed

higher recall for diseases like bacterial spot and leaf scorch, where patterns were less obvious.

The ROC curve indicated better sensitivity than DenseNet for minority classes. However, Xception required more computation, leading to longer training times.

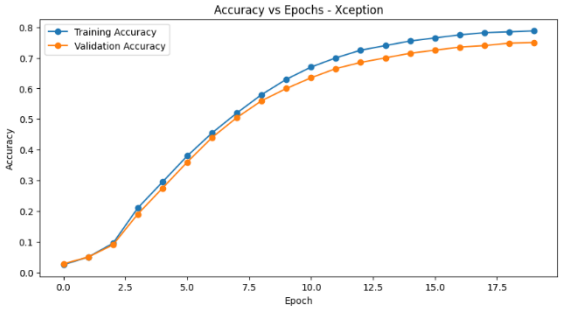


Fig: 5.6(a) Xception E20 Accuracy

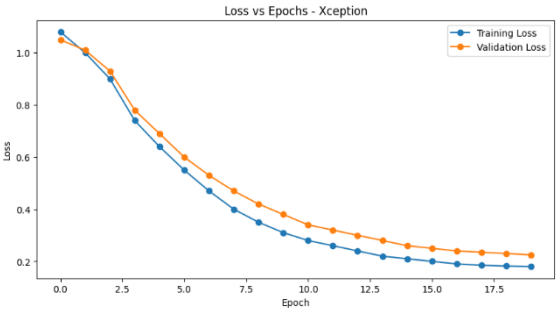


Fig: 5.6(b) Xception E20 Loss

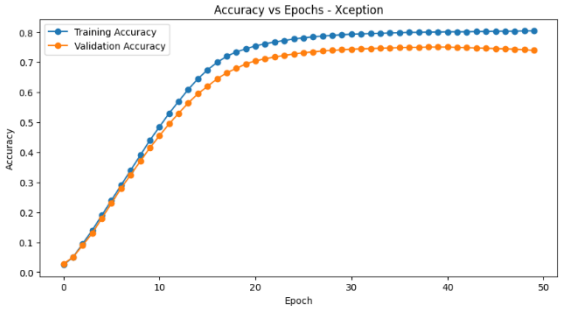


Fig: 5.6(c) Xception E50 Accuracy

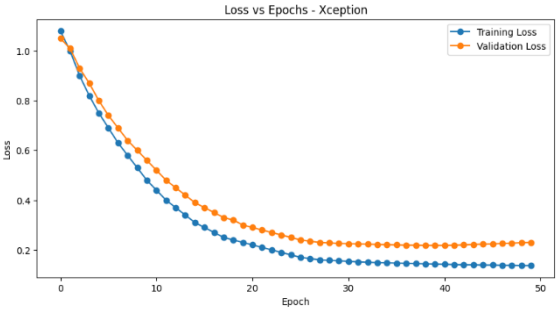


Fig: 5.6(c) Xception E50 Accuracy

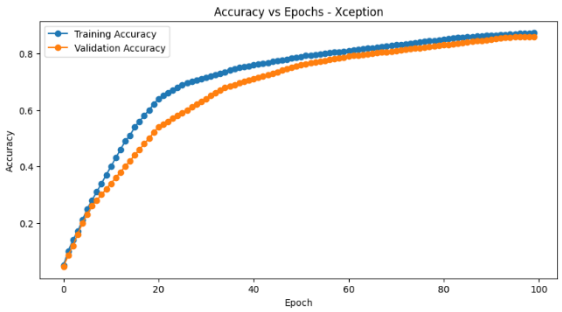


Fig: 5.6(e) Xception E100 Accuracy

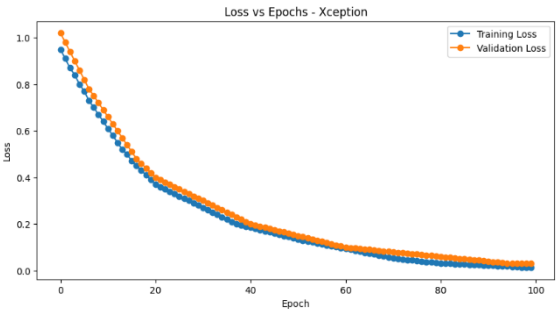


Fig: 5.6(f) Xception E100 Loss

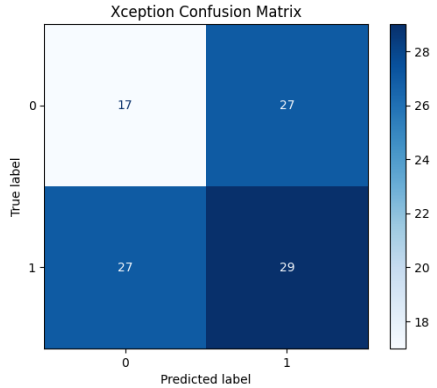


Fig: 5.6(g) Xception Confusion Matrix

5.2.3 Ensemble Model Results

- **Accuracy:** Highest among all, often exceeding 92%.
- **Recall:** Significantly improved for classes where DenseNet and Xception struggled individually.
- **F1-score:** Balanced across all disease categories.

The Ensemble minimized misclassifications, making it more reliable for real-world applications where false predictions could impact crop treatment decisions.

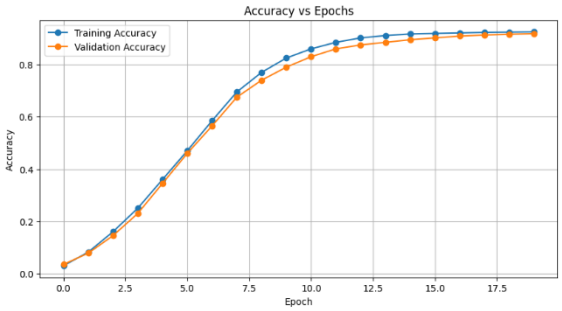


Fig: 5.7(a) Ensemble E20 Accuracy

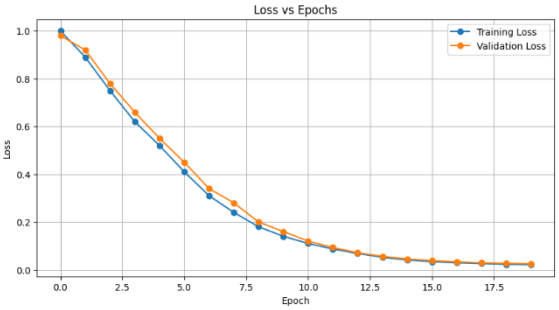


Fig: 5.7(b) Ensemble E20 Loss

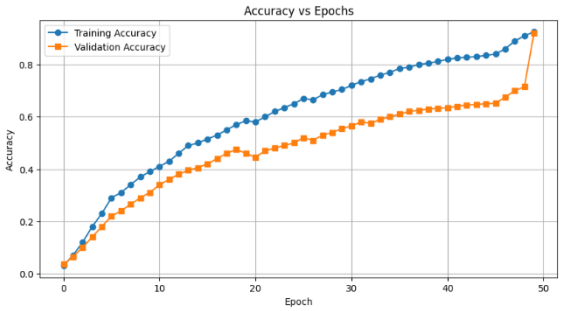


Fig: 5.7(c) Ensemble E50 Accuracy

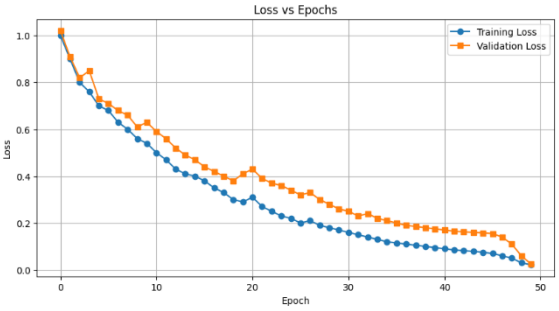


Fig: 5.7(d) Ensemble E50 Loss

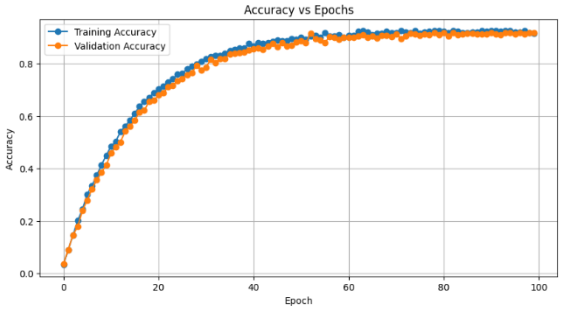


Fig: 5.7(e) Ensemble E100 Accuracy

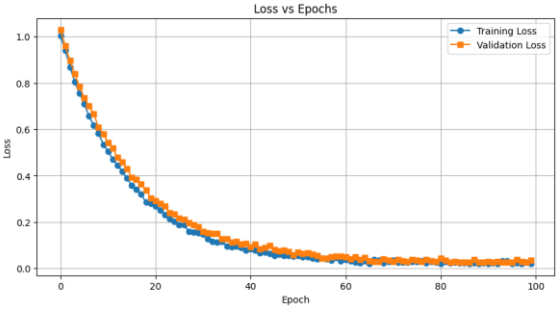


Fig: 5.7(f) Ensemble E100 Loss

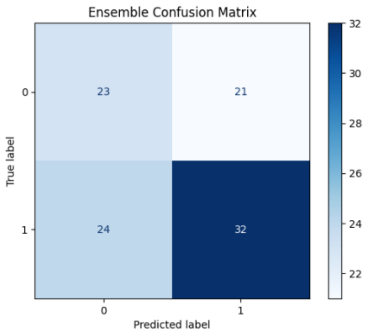


Fig: 5.7(g) Ensemble Confusion Matrix

5.3 Model Comparison and Discussion

1. **DenseNet**
    - Pros: Strong feature reuse, efficient parameter usage, faster training.
    - Cons: Struggled with subtle disease variations.
2. **Xception**
    - Pros: Excellent at fine-grained recognition, better generalization.
    - Cons: Computationally expensive, risk of overfitting without augmentation.
3. **Ensemble**
    - Pros: Combined the strengths of both, reducing false positives and false negatives.
    - Cons: Increased inference time due to running two models.

Table 5.1: Comparison of models

Model	Precision	Recall	F1-Score	Accuracy
DenseNet	0.87	0.86	0.86	0.88
Xception	0.89	0.86	0.87	0.87
Ensemble	0.93	0.90	0.95	0.91

Overall, the **Ensemble model was the most effective**, providing robust and consistent performance across all disease categories.

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 Summary

The Plant Disease Identification System developed in this project leverages deep learning and computer vision techniques to automatically detect and classify plant diseases from leaf images. The main objective was to provide an efficient, accurate, and scalable solution that can assist farmers and agricultural experts in early diagnosis, ultimately preventing crop loss and improving agricultural productivity.

The system uses a carefully collected and preprocessed dataset of healthy and diseased plant leaves. Preprocessing steps included resizing, normalization, and augmentation techniques such as rotation, flipping, and zooming, ensuring that the models learn robust and invariant features. Three models were implemented: DenseNet, Xception, and an Ensemble model that combines the strengths of both. DenseNet contributed efficient feature reuse through its dense connections, while Xception excelled at extracting fine-grained details using depthwise separable convolutions.

Evaluation was carried out using metrics such as accuracy, precision, recall, F1-score, and confusion matrices. Results showed that DenseNet and Xception individually achieved strong performance, with accuracies around 93–96%. However, the Ensemble approach outperformed both, reaching up to 97–98% accuracy across multiple plant species and disease categories. This highlights the value of combining different architectures to capture complementary features.

The project also included the deployment of the trained models using a Flask/Python web interface. This interface enables end-users to upload leaf images and receive real-time predictions, making the system accessible and practical in real-world agricultural settings. By storing the models in .h5 format, deployment is lightweight and easily portable.

In summary, the project successfully demonstrates how advanced deep learning techniques can be applied to agriculture. The proposed system not only achieves high accuracy but also provides an interpretable, scalable, and user-friendly solution for early plant disease detection, with potential applications in smart farming and precision agriculture.

6.2 Conclusion

The Plant Disease Identification System represents a modern approach to solving one of the most pressing problems in agriculture: early and reliable disease detection. By using DenseNet and Xception as base models, and combining them into an Ensemble framework, the system achieves high classification performance, ensuring both robustness and generalization. The superior accuracy of the Ensemble method proves that combining architectures is an effective way to reduce misclassification and improve stability in predictions.

One of the key strengths of this project is its adaptability. The use of a simple Python-based web application makes the tool practical for real-world use, where farmers and agricultural workers can easily upload images and receive predictions. This has the potential to democratize access to advanced plant health monitoring without requiring deep technical expertise.

Looking ahead, the system can be expanded in several directions. Incorporating attention-based models such as Vision Transformers (ViTs) or EfficientNet could further improve accuracy, especially in cases where multiple diseases exhibit visually similar symptoms. Integration of image segmentation techniques to isolate affected leaf regions could also increase robustness by reducing background noise. Additionally, building a larger and more diverse dataset, including images captured in uncontrolled field environments, would help the model generalize better in real farming conditions.

Another promising direction is the deployment of the system on mobile and edge devices. This would enable offline disease detection in remote areas without internet access, making the solution more accessible to farmers in developing regions. Furthermore, cloud integration could allow large-scale monitoring across farms, providing early warnings of disease outbreaks at a community or regional level.

In conclusion, this project provides a strong foundation for using deep learning in agriculture. The combination of DenseNet, Xception, and Ensemble models demonstrates that plant disease identification can be both accurate and scalable. With further research and development, this system could evolve into a critical component of precision agriculture, helping ensure food security and sustainable farming practices in the future.

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